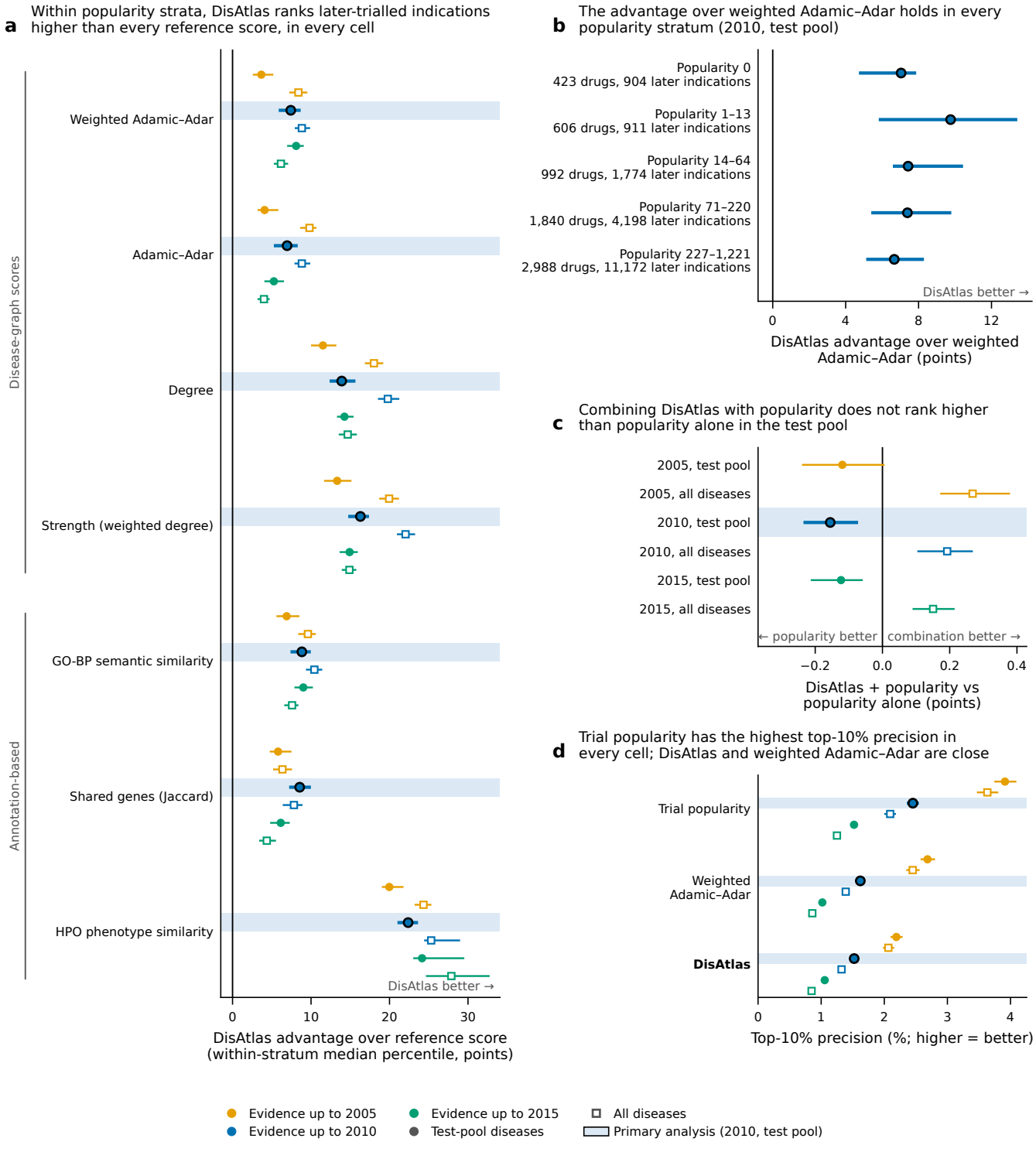
a
Within popularity strata, DisAtlas ranks later-trialled indications higher than every reference score, in every cell
Disease-graph scores
Annotation-based
Weighted Adamic–Adar
Adamic–Adar
Degree
Strength (weighted degree)
GO-BP semantic similarity
Shared genes (Jaccard)
HPO phenotype similarity
DisAtlas better →
0
10
20
30
DisAtlas advantage over reference score (within-stratum median percentile, points)
b
The advantage over weighted Adamic–Adar holds in every popularity stratum (2010, test pool)
Popularity 0
423 drugs, 904 later indications
Popularity 1–13
606 drugs, 911 later indications
Popularity 14–64
992 drugs, 1,774 later indications
Popularity 71–220
1,840 drugs, 4,198 later indications
Popularity 227–1,221
2,988 drugs, 11,172 later indications
DisAtlas better →
0
4
8
12
DisAtlas advantage over weighted Adamic–Adar (points)
c
Combining DisAtlas with popularity does not rank higher than popularity alone in the test pool
2005, test pool
2005, all diseases
2010, test pool
2010, all diseases
2015, test pool
2015, all diseases
← popularity better
combination better →
−0.2
0.0
0.2
0.4
DisAtlas + popularity vs popularity alone (points)
d
Trial popularity has the highest top-10% precision in every cell; DisAtlas and weighted Adamic–Adar are close
Trial popularity
Weighted Adamic–Adar
DisAtlas
0
1
2
3
4
Top-10% precision (%; higher = better)
Evidence up to 2005
Evidence up to 2010
Evidence up to 2015
Test-pool diseases
All diseases
Primary analysis (2010, test pool)